

CS390R: Project 2

February 28, 2023

Please submit all screenshots in one pdf file with each clearly labeled (The corresponding question) for grading. Code should be handed in as separate py files with the specified names. The writeup for challenge three should be its own pdf file. All other files will be ignored.

1. The first challenge is a simple buffer overflow where you have to jump through multiple functions in a certain order to pass checks. First, please list the location of the checks in the program. You can put this in your writeup. (30 pts). Then provide a python script called `Lastname_Firstname_p2c1.py` that runs your exploit and prints the flag (70 pts).
2. This challenge is a simple note taking app. First, please explain the error in the code that can lead to exploitation (20pts). Then, think about how you could exploit that code if the if notes was mmaped with a random address instead of the hardcoded one? (20pts). You can provide both those answers in your write up. Last, submit a python script called `Lastname_Firstname_p2c2.py` that runs your exploit and spawns a shell. Also provide a screenshot of the exploit script working on your machine (the shell running from it) (60pts).
3. Submit a `Lastname_Firstname_p2c3.py` file that exploits the program and runs shellcode that will open, read and write the `flag.txt` file contents back to you. Do not use shellcode to spawn a shell for this challenge, just use file io syscalls (60 pts). The writeup is worth (40 pts).